

Original Research

Seroprevalence of toxoplasmosis in antenatal women with bad obstetric history

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KEY WORDS

Antenatal women, assay, bad obstetric history, enzyme linked immunosorbent immunoglobulin G, intrauterine death, immunoglobulinM, toxoplasmosis

ABSTRACT

Background: The occurrence of fetal death is one of the tragedies that confront the physician providing obstetric care. Among the various agents associated with infections of pregnancy, viruses are the most important followed by bacteria and protozoa. Among protozoal infections in pregnancy, toxoplasmosis is reported to have a high incidence, sometimes causing fetal death. The study was intended to observe the seroprevalence of Toxoplasmosis in pregnant women presenting with bad obstetric history (BOH). **Materials and Methods:** A total of 92 antenatal women were included in the study (80 in the study group and 12 in control group). The study group comprised of antenatal women with BOH in the age group of 20-35 years. Antenatal women with Rh incompatibility, pregnancy induced hypertension, diabetes mellitus, renal disorders and syphilis were not included in the study. The control group included women in reproductive age group without BOH. All the samples were screened by enzyme linked immuno sorbent assay (ELISA) for *Toxoplasma* specific Immunoglobulin M (IgM) and Immunoglobulin G (IgG) antibodies. **Results:** Of the 80 antenatal women in the study group, 36 (45%) were seropositive for *Toxoplasma* specific IgG antibodies ($P < 0.005$), 16 (20%) were seropositive for *Toxoplasma* specific IgM antibodies ($P < 0.005$) and 8 (10%) were seropositive for both IgG and IgM antibodies ($P < 0.005$). Various predisposing factors for acquiring Toxoplasmosis such as contact with cats, contact with soil, food habits, illiteracy, socio-economic status and residential status were also studied. **Conclusions:** We conclude that toxoplasmosis during pregnancy causes congenital fetal infection with possible fetal loss. ELISA was found to be a sensitive serological test for diagnosis of Toxoplasmosis in pregnant women with BOH. Major cause of fetal loss in BOH cases in the study group was abortion.

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INTRODUCTION

Toxoplasmosis is a zoonotic protozoal infection. It occurs during pregnancy as an acute infection that may cause damage to the fetus, which is one of the important reasons for bad obstetric history (BOH) in pregnant women.^[1] *Toxoplasma* infection is known to be transmitted from cat's feces, often through contaminated vegetables, fruits and milk.

Diagnosis of *Toxoplasma* specific Immunoglobulin M (IgM) by enzyme linked immuno sorbent assay (ELISA) indicates recent infection as the cause of BOH in pregnant women leading to congenital malformations, abortions

and still births. The detection of *Toxoplasma* specific immunoglobulin G (IgG) antibodies indicates chronic infection.^[2]

Studies analyzing the role of maternal infection in the causation of BOH are less in number, probably due to lack of facilities in isolating etiological agents causing BOH and the prohibitive cost of commercial diagnostic kits. Most of the available information on Toxoplasmosis from India is in women with pregnancy wastage.

In view of the above observation, the present study was carried out on clinically suspected cases of toxoplasmosis in antenatal women. The study was intended to observe the sero prevalence of toxoplasmosis in pregnant women with BOH.

MATERIALS AND METHODS

A total of 92 whole blood samples were collected from antenatal women attending their antenatal check-up. They were divided into two groups, namely the study group and the control group.

Study group

Inclusion criteria

Women in reproductive age group with BOH such as; spontaneous abortions, missed abortions, intra uterine death (IUD), preterm deliveries, still births, congenital malformations, perinatal deaths, unexplained neonatal deaths, women having mentally retarded children in earlier pregnancies and unexplained seizures.

Exclusion criteria

Patients with Rh incompatibility, pregnancy induced hypertension, diabetes mellitus, renal disorders and syphilis.

Control group

Women in reproductive age group without any BOH. Every case was studied by obtaining detailed history and data regarding clinical examination, history of lymphadenopathy, pyrexia of unknown origin, convulsions and visual disturbances which indicate symptomatic toxoplasmosis, was specifically elicited. Subsequently relevant investigations to exclude common etiological factors implicated in BOH were conducted which included blood grouping, Rh typing, complete hemogram, coagulation profile, fasting and post-prandial blood sugars and renal function tests. Thus only idiopathic fetal wastage was included in the present study.

Methods of sample collection

A total of 92 whole blood samples were collected using sterile disposable syringes. Under aseptic precautions, 5ml of blood was withdrawn by venepuncture. After

centrifugation, clean serum was transferred into provials and was preserved at -20°C .

Processing of serum samples

All the samples were screened for *Toxoplasma* specific IgG and IgM antibodies by ELISA. (EUROIMMUN IgM and IgG) manufactured by Medizinische Labordiagnostica, Germany.

Test protocol

The antigen coated is lysate of sonicated gamma irradiated tachyzoites. Patient's serum was added and any antibodies to *Toxoplasma* antigens present in the serum would bind to these antigens. The microwells were washed to remove unbound serum proteins. Antibodies conjugated with Horseradish Peroxidase enzyme and directed against human IgM and IgG were added which would in turn bind to any human IgM and IgG present. The microwells were washed to remove unbound conjugate, and then chromogen/substrate was added. In the presence of peroxidase enzyme the colorless substrate was hydrolyzed to a colored end product. The color intensity was proportional to the amount of antibodies present in the patient serum.

Measurements

Photometric measurement of the color intensity was measured using 450 nm filter within 30 min by adding the stop solution.

Calculation of results

The ratios were calculated according to the following formula.

$$\frac{\text{Extinction value of the control or patient sample}}{\text{Extinction value of calibrator}} = \text{Ratio}$$

Interpretations of results are as follows:

Ratio < 8.0: Negative

Ratio < 8.0-1.0: Borderline

Ratio > 1.0: Positive

A positive result indicates presence of IgM or IgG antibodies to *Toxoplasma gondii*.

A negative result indicates no immunity to *Toxoplasma*. These individuals are presumed to be susceptible to a primary infection.

The same procedure was also repeated for IgG antibodies.

RESULTS

Out of 80 antenatal women in the study group 36 (45%), were seropositive for *Toxoplasma* specific IgG ($P < 0.005$), 16 cases (20%) were seropositive for *Toxoplasma* specific

IgM antibodies ($P < 0.005$) and 8 cases (10%) were seropositive for both IgG and IgM ($P < 0.005$). In the control group, 1 sample each was seropositive for IgM and IgG antibodies [Table 1].

The majority of seropositive cases in the study group were in the age group 20-24 years (49.99%). Among the seropositive cases IgG seropositivity was more in 20-24 years as compared 30-34 years [Table 2].

IgG seropositivity was high among antenatal women of low socioeconomic status (83.87%) ($P < 0.05$), illiterates (80%) ($P > 0.05$), from rural area (78.12%) with non-vegetarian food habits (72.34%) ($P < 0.05$). IgM seropositivity was high among antenatal women with high socioeconomic status (80%) ($P < 0.05$), with the history of contact with cats (60%) ($P < 0.05$) and with the history of contact with soil (38.7%) ($P > 0.05$) [Table 3].

There was gradual increase of seropositivity for *Toxoplasma* infection with increasing parity for both IgG and IgM antibodies. These roprevalence was more in Gravida VI (87.5%) for IgG and (62.5%) for IgM antibodies [Table 4].

The predominant clinical condition responsible for BOH with *Toxoplasma* infection was abortions in 20 cases (38.46%) followed by IUD (24.99%), still births (28.5%), hydrocephalus and anencephalus [Table 5].

Ten seropositive cases in the present study were treated with spiramycin. Six of them were given prolonged therapy and four of them treated with interval therapy. Out of these 10 cases eight gave birth to live children [Table 6].

DISCUSSION

Increasing evidence is available which shows that infection by *Toxoplasma* induces foetal loss in women. This agent has predilection for nucleated cells of muscle, intestinal epithelium and placenta. It can be congenitally acquired by transfer through the placenta, if the mother contracts the disease during pregnancy, causing abortions, stillbirths, congenital malformations.^[3] The disease can be diagnosed by serology. Demonstration of *Toxoplasma* specific IgM antibodies indicates recent infection and detection of *Toxoplasma* specific IgG antibodies indicates chronic infection.

In this study out of 80 cases in the study group 16 (20%) were seropositive for *Toxoplasma* specific IgM antibodies, 36 cases (45%) were seropositive for *Toxoplasma* specific IgG antibodies. In Andhra Pradesh, the results of the present study are comparable to the observations made by Dr. Sankar 2004, who reported

Table 1: Correlation between IgG and IgM seropositivity among bad obstetric history and control groups ($n=92$)

Result	BOH cases		Normal controls		P value
	No. of cases	%	No. of cases	%	
Only IgG positive	28	35	1	8.33	<0.005
Only IgM positive	8	10	1	8.33	<0.005
IgG and IgM positive	8	10	-	0	<0.005
IgG and IgM negative	36	45	10	83.33	0.001
Total	80	100	12	100	

IgM: Immunoglobulin M, IgG: Immunoglobulin G, BOH: Bad obstetric history

Table 2: Seroprevalence of *toxoplasma* immunoglobulin G and immunoglobulin M in different age groups among bad obstetric history cases ($n=80$)

Age group in years	Total no. of cases	IgG positive (%)	IgM positive (%)
15-19	0	0	0
20-24	26	16 (61.53)	10 (38.46)
25-29	19	14 (73.68)	5 (26.31)
30-34	7	6 (85.71)	1 (14.28)
Total	52	36	16

IgM: Immunoglobulin M, IgG: Immunoglobulin G

Table 3: Source of infection among seropositive cases ($n=52$)

Status	Total no. of cases	IgG positive (%)	IgM positive (%)
Contact with cats			
Present	20	8 (40)	12 (60) ($P < 0.05$)
Absent	32	28 (87.5)	4 (12.5)
Contact with soil			
Present	31	19 (61.29)	12 (38.7) ($P > 0.05$)
Absent	21	17 (80.95)	4 (19.04)
Food habits			
Vegetarian	5	2 (40)	3 (60) ($P < 0.05$)
Non-vegetarian	47	34 (72.34)	13 (27.65)
Socioeconomic status			
Lower	31	26 (83.8)	5 (16.1)
Middle	16	9 (56.2)	7 (43.7)
Higher	5	1 (20)	4 (80)

($P < 0.05$), IgM: Immunoglobulin M, IgG: Immunoglobulin G

Table 4: Analysis of results in relation to parity ($n=80$)

Gravida	Seropositive			Seronegative	
	No. of cases	IgG positive No. of cases (%)	IgM positive No. of cases (%)	No. of cases	%
III	21	18 (85.71)	3 (14.28)	10	35.71
IV	2	8 (66.66)	4 (33.33)	9	32.14
V	11	7 (63.66)	4 (36.36)	5	17.85
VI	8	7 (87.5)	5 (62.5)	4	14.28
Total	52	36	16	28	

($P < 0.05$), IgM: Immunoglobulin M, IgG: Immunoglobulin G

a seropositivity of 38% for IgG and Yasodhara *et al.*, who reported a seropositivity of 18.3% for *Toxoplasma*

specific IgM antibodies.^[4] Table 5 summarizes similar studies which have determined the seropositivity for *Toxoplasma* specific antibodies. In the present study the values are comparatively higher when compared with other studies done by indirect fluorescent antibody test and indirect haemagglutination. The high seropositivity rate obtained in the present study can be attributed to the high sensitivity of the test system used to detect *Toxoplasma* antibodies.^[5]

Out of 80 cases in the study group, eight cases (10%) were seropositive for both IgG and IgM antibodies with possible recent infection in the last 12 months. Eight cases (10%) were seropositive for IgM antibodies alone, which indicates acute infection. 28 cases (35%) were seropositive for IgG antibodies only which indicates chronic infection. Thirty six cases (45%) were seronegative for both IgG and IgM antibodies which shows that the BOH was not due to *Toxoplasma* infection, but may be due to other infections like CMV (cytomegalovirus), Herpes simplex and Rubella.^[6]

Out of 12 cases in the Control group, one was seropositive for IgG antibodies which may be due to exposure to

Toxoplasma infection in the past. Also, one patient in the control group was positive for IgM antibodies which probably denote a subclinical infection.

Out of 52 seropositive cases in the study group the highest incidence of toxoplasmosis was observed in the age group 20-24 years, 26 cases (49.99%). It was observed that the seropositivity for *Toxoplasma* specific IgG antibodies increased from the age group 20-24 years 61.53% (16 out of 26) to 85.71% (6 out of 7) in 30-34 year age group. Mittal *et al.*, in their study revealed that the lowest rate of antibody acquisition occurred in the age group 15-20 years and it was increased with increase in age profile.^[7] This is in agreement with the present study.

In our study abortions constituted the major clinical case of pregnancy wastage when compared to still births, IUD, congenital malformations like hydrocephalus, anencephaly.^[8] The second major presenting condition was IUD, 12 cases (23.07%) of which nine (74.99%) were seropositive for IgG and three were seropositive for IgM antibodies. There were seven still births (13.46%) and 10 preterm labour cases (19.23%).

IgG seropositivity was more in still births, IgM seropositivity was more in preterm labour cases.^[9]

There is evidence that active *Toxoplasma* infection can occasionally persist from one pregnancy to the next and such infection may be the cause repeated miscarriages or still births.^[10] It appears that in such cases, *T. gondii* may encyst in the uterine endometrium and is stirred into activity by the process of placentation. This may lead to low grade local endometritis which may persist. *T. gondii* has been isolated from abortus in one case of

Table 5: Clinical spectrum of seropositive bad obstetric history cases (n=52)

Clinical conditions	Total no of cases (%)	IgG positive (%)	IgM positive (%)
Abortions	20 (38.46)	15 (75)	5 (25)
Still births	7 (13.46)	5 (71.42)	2 (28.5)
IUD	12 (23.07)	9 (74.99)	3 (24.99)
Hydrocephalus	2 (3.8)	-	2 (100)
Anencephaly	1 (1.9)	1 (100)	-
Preterm labor	10 (19.23)	6 (60)	4 (40)
Total	52	36	16

IgM: Immunoglobulin M, IgG: Immunoglobulin G, IUD: Intrauterine death

Table 6: Percentage seropositivity of toxoplasmosis in different study groups in India

Year	Place	Author	No. of sera tested	Method	IgG positive (%)	IgM positive (%)
1994	Delhi	Raghunadhan <i>et al.</i>	50	IgMELISA	-	12
1994	Amritsar	Sandhu <i>et al.</i>	100	IHA	22	5
1995	Solapur	Kandle <i>et al.</i>	84	LAT	50	-
1995	Delhi	Mittal <i>et al.</i>	2075	IFAT	7.71	-
1996	Visakhapatnam	Anuradha	80	EIA	35	-
2001	Hyderabad	Yasodhara <i>et al.</i>	175	ELISA IgM	-	13.1
2001	Kakinada	Sankar	225	ELISA IgG	38	-
2001	Warangal	B. Anuradha	100	ELISA IgM	-	48
2003	Mumbai	D. Turbadkar	380	Sandwich ELISA	42.1	10.5
2003	Maharashtra	-	194	ELISA	35	15
2003	Delhi	Rawal	100	ELISA	41.2	-
2004	Hyderabad	Yasodhara	100	ELISA	33	18.3
2004	Delhi	Singh S	180	ELISA	45	3.3
2004	Guntur	Hemaprakashkumari	80	ELISA	28.75	50
2005	Visakhapatnam	Present study	80	ELISA	45	20

IgM: Immunoglobulin M, IgG: Immunoglobulin G, LAT: Latex agglutination test, IHA: Indirect haemagglutination test, IFAT: Indirect fluorescent antibody test, EIA: Enzyme immuno assay, ELISA: Enzyme linked immuno sorbent assay

miscarriage. Hence it is essential to test for both IgG and IgM antibodies, as chronic infection can also lead to fetal wastage.

From this study it is observed that illiteracy, poverty, overcrowding, lack of hygiene and associated environmental factors play a crucial role for the high incidence of Toxoplasmosis. The study also shows that, women who had contact with the soil ($P < 0.005$) and those with non-vegetarian food habits have high levels of IgG antibodies ($P < 0.005$).^[11] This shows that unique environmental factors in various communities, eating habits have s on the transmission of this infection. The socio-epidemiological aspects of toxoplasmosis are the important contributing factors for the spread of the disease.

Early diagnosis and treatment have an effective role in reducing transmission of infection from mother to baby.^[12] Ten seropositive cases in the present study were treated with Spiramycin. Four of them were treated with interval therapy (Spiramycin 3 million IU International units twice a day for 3 weeks) and six of them with prolonged therapy during pregnancy (Spiramycin 3 million IU twice a day for 3 weeks, with 2 weeks drug free period) same treatment was continued till parturition. Outcome of these cases was excellent after treatment with Spiramycin. Of the 10 cases, 80% had healthy live children.^[13-15]

Toxoplasmosis is amenable to treatment. Early detection with repeated serological examination and treatment in all pregnancies can reduce the hazard substantially.

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